Quenching constants of KrF(B, C) by Kr and Xe and the KrF(B, C) equilibrium constant

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The two-photon, photo-assisted reaction between Kr and F_2 has been used to prepare KrF(B, C) molecules with a 7 ns laser pulse for subsequent decay measurement in Kr and Xe/Kr mixtures. The two- and three-body quenching constants in Kr are $(4.0\pm1.0)\times10^{-12}$ cm^3 s^{-1} and $(6.0\pm2.0)\times10^{-31}$ cm^6 s^{-1}, respectively. The KrF(B, C) equilibrium constant at 300 K is $\approx$4–5 with KrF(C) as the lower energy state. Quenching of KrF(B, C) by Xe gives XeF(B, C), and possibly some XeF(D), as a product with a rate constant of $(10\pm2)\times10^{-11}$ cm^3 s^{-1}.

1. Introduction

Most of the KrF(B, C) quenching constants available in the literature were measured under less than ideal conditions [1–7]. The degree of coupling between the KrF(B) and KrF(C) states often was not specified, which can make proper assignment of the observed decay rates difficult and deconvolution of the observed decay rates into radiative decay rates and heavy-body quenching rates imprecise. In fact, since the equilibrium constant for KrF(B) and KrF(C) is not well known [3,8], the effective radiative decay time for the collisionally coupled levels is still not established. We wish to report new decay constant measurements for KrF(B, C) under fully coupled 300 K Boltzmann conditions [9,10] in Kr and in Xe/Kr mixtures. The two-photon, photo-assisted reaction [11] between Kr and F_2 was used to prepare KrF(B, C) molecules with a 7 ns laser pulse

$$\mathrm{Kr}+\mathrm{F_2}+2h\nu\rightarrow\mathrm{KrF(B,C)}+\mathrm{F}\,. \qquad (1)$$

The KrF(B, C) molecules were prepared in sufficiently high pressure of Kr bath gas ($\gtrsim$30 Torr) to ensure that vibrational and electronic state relaxation to 300 K distributions occurs before the KrF(B, C) decay measurements are made. This fast preparation method of KrF(B, C) has the additional advantage that other gases can be added to the Kr/F_2 mixture without interfering with the generation of KrF(B, C). The experimental details of the method, including a full presentation of the two-state coupled model, has been described in a similar study of XeCl(B, C) quenching [12].

In the present work, the KrF(B, C) decay rates (monitored by the KrF(B–X) fluorescence) in the presence of 50–500 Torr of Kr were measured to establish the coupled radiative decay lifetime, the two-body quenching rate constant and the three-body quenching rate constant

$$\mathrm{KrF(B,C)}+\mathrm{Kr}\rightarrow2\mathrm{Kr}+\mathrm{F}\,, \qquad (2)$$

$$\mathrm{KrF(B,C)}+2\mathrm{Kr}\rightarrow\mathrm{Kr_2F^*}+\mathrm{Kr}\,. \qquad (3)$$

Fortunately, the KrF(B, C) coupled radiative lifetime at 300 K is sufficiently long, $\approx$25 ns, so that a significant reduction in the decay time with added Kr could be observed. The method used to generate KrF(B, C) is not sufficiently state specific to be suitable for studying B$\leftrightarrow$C mixing rates. Experiments also were done with Xe added to a F_2/Kr mixture (1.5 Torr of F_2 in 100–150 Torr of Kr) to isolate and observe the quenching of KrF(B, C) by Xe

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$$KrF(B, C) + Xe \rightarrow XeF(D, B, C) + Kr\,, \quad (4a)$$

$$KrF(B, C) + Xe \rightarrow Xe + Kr + F\,. \quad (4b)$$

The direct observation of XeF(B, C) formation in (4a) indicates that the displacement channel dominates over the physical quenching channel. Previously [11] we reported the displacement reaction between Kr_2F^* and Xe using the same general experimental approach as used here; however, those experiments were done with 500 Torr of Kr so that the KrF(B, C) molecules were converted to Kr_2F^* molecules before encountering a Xe atom.

2. Experimental methods

The KrF(B, C) molecules were generated by a 7 ns (fwhm) dye laser pulse at 240–242 nm in a static, laser-induced fluorescence cell constructed from stainless steel and fitted with baffle arms to reduce scattered laser light. The laser beam was focused in the center of the cell with a 0.5 m lens. The dye laser (Quanta Ray PDL-1) was pumped by a frequency doubled Nd:YAG laser (Quanta Ray DCR). The 240–242 nm wavelength range was generated from mixing the second harmonic of the dye laser (DCM dye) with the fundamental of the Nd:YAG laser in a KD*P crystal. The laser pulse energy was in the range of ≈ 1 mJ/pulse. The F_2/Kr excitation spectrum is broad (235–255 nm) and the specific excitation wavelength is not very important, except that scattered laser light must be avoided in observing the KrF(B–X) fluorescence.

The KrF(B–X) fluorescence was collected with a short focal length lens and focused onto the entrance slit of a 1 m monochromator set at 248 nm. The signal was detected with a R-2693 (Hamamatsu) photomultiplier tube mounted in a base that was wired for fast response; the rise time of the photomultiplier tube was 1.5 ns. The fluorescence signal was recorded and accumulated with either a LeCroy Digitizer (1.3 G sample/s) or with a Hewlett Packard digital storage scope (1.0 G sample/s). All the KrF(B–X) waveforms were single exponential for pressures above 50 Torr of Kr. At low pressure ($\lesssim 10$ Torr) the KrF(B) and KrF(C) states are not fully coupled and two (or more) decay constants can be observed. Since the decay time of the laser pulse was about 4 ns, KrF(B, C) decay times of $\geqslant 5$ ns could be reliably observed by following the fluorescence signals for more than two decay time periods. Since the KrF(B–X) fluorescence was quite strong for high pressures of Kr, observing the fluorescence for such time periods was not difficult.

The F_2 was used as a 10% mixture in He purchased from Matheson Gases. The F_2 composition was confirmed by making duplicate KrF(B, C) measurements [13] with another F_2/He mixture that had been checked for composition by comparison with pure F_2 [11,13]. The Kr and Xe were research grade purchased from Cryogenic Rare Gases. The gas cell and gas handling lines were passivated with F_2 prior to use and periodically thereafter, if the KrF(B–X) fluorescence intensity showed any diminution over a 30 min period of irradiation. Pressures were measured with MKS Baratron pressure transducers, either 0–10 Torr or 0–1000 Torr heads, which were calibrated with a Hg manometer.

3. Results and discussion

The two-photon photo-assisted reaction of Kr/F_2 has a smaller cross section [13] than that of Xe/Cl_2, and higher pressures of F_2 must be used to generate acceptable concentrations of KrF(B, C) than for Cl_2 in the generation XeCl(B, C) [12] from Xe/Cl_2 mixtures. Therefore, an accurate measurement of the KrF(B, C) quenching constant by F_2 was needed. Two series of experiments were done with constant Kr pressure of 50 and 100 Torr and added F_2. The KrF(B, C) decay constants are plotted versus F_2 pressure in fig. 1. Both series of experiments gave the same value for the F_2 quenching constant and the average value is $(2.8 \pm 0.2) \times 10^{-10}$ cm^3 s^{-1}. This rate constant also agrees with an independent measurement in our laboratory with a different F_2 source [13]. Our rate constant is somewhat smaller than the value reported [5] for KrF(B) ($(4.8 \pm 0.3) \times 10^{-10}$ cm^3 s^{-1}) from photolysis of KrF_2 in variable pressure of F_2 and that for KrF(B, C) ($(5.9 \pm 0.5) \times 10^{-10}$ cm^3 s^{-1}) from excitation of Kr($^3P_1$) in F_2 reported by Quigley and Hughes [4]. All of these values are lower than the value favored in tabulations of rate constants used for modeling the

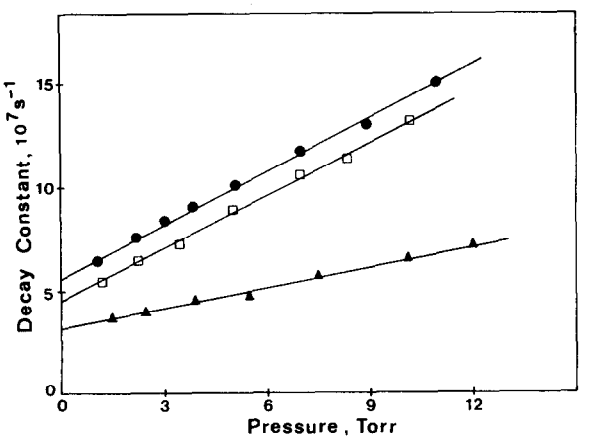


Fig. 1. Decay constants of KrF(B, C) for various pressures of F_2 in constant pressure of Kr; (●) 100 Torr Kr, (□) 50 Torr Kr and the decay constants for KrF(B, C) in Xe(▲) for 1.5 Torr of F_2 and 100 Torr of Kr. The pressure scale should be multiplied by 2 when reading the Xe data.

KrF laser [2,3], which was based upon deconvoluting the product of τ_{eff} and k_{F_2} obtained from electron beam experiments. The most uncertain quantity in our experiments (and others) is the exact $[F_2]$.

A series of experiments were done with constant $[F_2]$ and variable Kr pressures from 50–500 Torr in order to study Kr quenching. The results are summarized in fig. 2. The decay constants were always taken from the latest part of the KrF(B, C) waveform that could be analyzed. The decay times ranged from 25 to 4 ns; the highest Kr pressure ($\approx$500 Torr) that we could investigate was limited by the decay time of our laser pulse. Two things are self-evident even without detailed analysis: (i) the effective radiative decay time ($\tau_{eff}^{-1} = (\tau_B^{-1} + K_{eq}\tau_C^{-1})/(1+K_{eq})$), must have a large KrF(C) component in order to be $\approx$25 ns, since $\tau_B = 7.0$ ns and $\tau_C = 75$ ns [14] and (ii) the three-body quenching rate constant for KrF(B, C) must be smaller than for XeCl(B, C), because the XeCl(B, C) decay rates were to fast to be measured in 300 Torr of Xe.

The results in fig. 2 were fitted to the exact form of the second (slow) decay constant for a two-level coupled kinetic scheme [12]

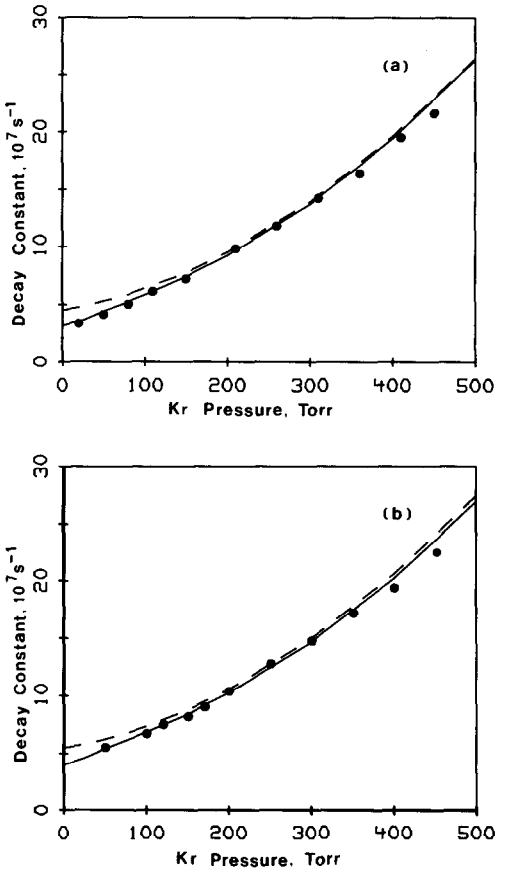


Fig. 2. Decay constants of KrF(B, C) for various pressures of Kr in 1 Torr of F_2 (a) and in 2 Torr of F_2 (b); the calculated values for λ^- (——) and λ_{ap}^- (- - -) are for $k_{F_2} = 3.0\times10^{-10}$ cm^3 s^{-1}, $k_{Kr} = 4.0\times10^{-12}$ cm^3 s^{-1}, $k'_{Kr} = 6.0\times10^{-31}$ cm^6 s^{-1} and $K_{eq} = 5.0$.

$$\lambda^- = \tfrac{1}{2}\{(K_1+K_2+K_{-2}+K_3) - [(K_1+K_2-K_{-2}-K_3)^2+4K_2K_{-2}]^{1/2}\},$$

$$K_1 = \tau_B^{-1} + k_{Kr}[Kr] + k'_{Kr}[Kr]^2 + k_{F_2}[F_2],$$

$$K_3 = \tau_C^{-1} + k_{Kr}[Kr] + k'_{Kr}[Kr]^2 + k_{F_2}[F_2],$$

$$K_2 = k_{BC}[Kr], \qquad K_{-2} = k_{CB}[Kr]. \tag{5}$$

All of the rate constants for KrF(B) and KrF(C) were assumed to be equal, except for the radiative decay constants ($\tau_B = 7.0$ ns and $\tau_C = 75$ ns) [14]. The ratio of k_{BC}/k_{CB} is K_{eq}; k_{BC} was assumed to be 1.0×10^{-10} cm^3 s^{-1} by analogy with k_{BC} for Xe with XeCl(B) [12] and K_{eq} was adjusted to get the best fit to the low pressure data of fig. 2. The exact value

for k_{BC} is not of crucial importance, although the interplay between $k_{BC}[Kr]$ and τ_B^{-1} determines the departure of λ^- from linearity in the lower pressure region. We used the calculated values [14] of τ_B and τ_C, which should be fairly reliable, as judged by the comparison between such calculated values and experimental measurements for KrF(B) [5] and for XeF and XeCl [1]. The best fit to the 1 and 2 Torr F_2 data were with $K_{eq}=5.0\pm1.0$, $k_{Kr}=4.0\times10^{-12}$ $cm^3\ s^{-1}$ and $k'_{Kr}=6.0\times10^{-31}$ $cm^6\ s^{-1}$. The quality of the fit to the 1 and 2 Torr F_2 data for the same set of constants is quite reasonable. Also, the rate of growth of Kr_2F^* with increase in [Kr] was consistent with the increase in three-body quenching rate, and we have no evidence to suggest a dark, three-body quenching component associated with reaction (3). The exact decay constant, λ^-, also can be compared with the approximate value, λ_{ap}^-,

$$\lambda_{ap}^- = \frac{\tau_B^{-1}+K_{eq}\tau_C^{-1}}{1+K_{eq}} + k_{F_2}[F_2]+k_{Kr}[Kr] + k'_{Kr}[Kr]^2, \quad (6)$$

which agrees with the exact λ^- value for Kr pressures >200 Torr. Extrapolation of (6) to low pressure gives a definition of the effective radiative lifetime, which was assigned as 25 ns after $k_{F_2}[F_2]$ was subtracted from the intercepts of fig. 2.

Our three-body quenching rate constant agrees moderately well with the value ($(2.9\pm0.3)\times10^{-31}$ $cm^6\ s^{-1}$) reported by Quigley and Hughes [4], which was obtained from the Kr dependence of the KrF(B, C) decay time at high pressure following the excitation of $Kr(^3P_1)$ in F_2. However, this agreement may be partly fortuitous, because their k_{F_2} value is larger than our result. Eden and co-workers [5] obtained k_{Kr} and k'_{Kr} values of $(8.6\pm0.7)\times10^{-12}$ $cm^3\ s^{-1}$ and $(9.7\pm0.8)\times10^{-31}$ $cm^6\ s^{-1}$ from photolysis of KrF_2 in 20–200 Torr of Kr; there is moderately good agreement, considering their narrow pressure range, but the intercept of their [Kr] plot is much different, probably because the photolysis of KrF_2 initially gives only KrF(B). If our coupled radiative lifetime value is used with the decay rate data of refs. [6,7], their k_{F_2} and k'_{Kr} values become $\approx2.0\times10^{-10}$ $cm^3\ s^{-1}$ and $\approx1.8\times10^{-31}$ $cm^6\ s^{-1}$, respectively. The two-body quenching rate by Kr was too slow to measure in that work. The more directly measured rate constants obtained in the present work should be the best results available to date for k_{Kr} and k'_{Kr} at 300 K.

These results for KrF(B, C) in Kr can be compared with the decay constants of XeCl(B, C) in Xe [12]. The two-body and three-body constants for Xe were $(5.0\pm2.5)\times10^{-12}$ $cm^3\ s^{-1}$ and $(13.0\pm4.0)\times10^{-31}$ $cm^6\ s^{-1}$, respectively. Although the two-body rate constants are nearly identical, the three-body constant is more than two times larger than that for XeCl(B, C) in Xe. This is probably related to the larger polarizability of Xe, which leads to greater attraction for the Xe/XeCl* potentials than for the Kr/KrF* potentials. The XeCl(B, C) equilibrium constant at 300 K is only 1.55 [15], and the energy difference between the B and C states must be larger for KrF* than for XeCl*. The interpretation of our KrF(B, C) data, which is dependent on the values for τ_B and τ_C, suggest an energy difference of ≈300 cm^{-1}; this value is two times larger than the ΔE_{BC} reported [8] from the high pressure extrapolation of the I_B/I_C steady-state intensity ratio in Ar of CF_4 bath gases. Neither the high pressure extrapolated intensity ratio [8] nor the low pressure extrapolation of λ_{ap}^- in fig. 2 are a precise measurement of ΔE_{BC}, but the collective data favor an energy separation of >200 cm^{-1} between KrF(B) and KrF(C).

The final set of measurements involved observation of the change in the KrF(B, C) decay time as a function of added Xe. The experiments were done with 1.5 Torr of F_2 and 100 Torr of Kr. Higher Kr pressures were not used in order to avoid complication from the presence of Kr_2F^*. The combined Xe/F_2 concentration was sufficiently low that the two-photon, photo-assisted reaction [13] giving XeF(B, C) was not important relative to reaction (4). The quenching data are shown in fig. 1; note that the value of the intercept agrees with the decay constant for fig. 2a at 100 Torr of Kr. The slope of the line in fig. 3 gives a two-body quenching constant of KrF(B, C) by Xe of $(10\pm2)\times10^{-11}$ $cm^3\ s^{-1}$. This large value is indicative of a different mechanism than the two-body quenching of KrF(B, C) by Kr in reaction (2). Fig. 3 shows the waveform for the decay of KrF(B, C) together with the delayed formation and subsequent decay of XeF(B), thus confirming the existence of the long accepted [1–3], but not definitively proven, displacement reaction of a Xe atom for a Kr atom in KrF(B, C) [16–18]. The de-

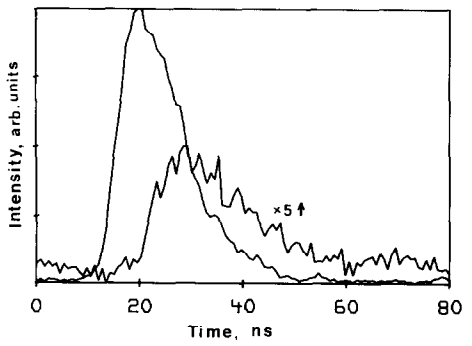


Fig. 3. Plot of the decay rate of KrF(B, C), the early waveform, and the growth of XeF(B), the late waveform, for an experiment in 150 Torr of Kr and 1.5 Torr of F_2 with 15 Torr of Xe. This Xe pressure corresponds approximately to half-quenching of KrF(B, C). The XeF(B) waveform was monitored at 350 nm; a 300 nm cut-off filter was placed between the cell and monochromator to eliminate Raleigh scattering of the laser light. If the concentration of XeF(C) is added to XeF(B), the relative intensity observed for XeF(B) is consistent with the degree of quenching for KrF(B, C).

lay in XeF(B) formation, relative to the 7 ns laser pulse, proves that XeF(B, C) is not being generated by the two-photon, photo-assisted reaction with Xe/F_2 [13]. Since collisions with Kr will promote transfer between XeF(B) and XeF(C), no information about the selectivity for XeF(B) versus XeF(C) formation in reaction (4) can be obtained from our experiments. We also attempted to observe XeF(D) by monitoring the XeF(D–X) fluorescence at 265 nm. Unfortunately, the KrF(C–A) emission was sufficiently strong that conclusive evidence about XeF(D) formation could not be obtained. In fact, the emission intensity observed at 265 nm was actually less in the presence of 15 Torr of Xe than without added Xe. The very fast quenching rate of KrF(B) by Xe observed by Eden et al. [18] from the photolysis of KrF_2 probably represents the collisional transfer from KrF(B) to KrF(C). The majority of reaction (4) proceeds to give XeF(C, B), but we cannot exclude a contribution from (4b) and some formation of XeF(D). Reaction (4a) proceeds via the bound, ground state KrXeF* potential reported by Apkarian and co-workers [19]. The endoergic reverse displacement reaction can be driven by vibrational energy in XeF(B, C) [16]. Since XeF(B, C) also has a significant two-body quenching rate by Kr [17], the description of the Kr–Xe–F potentials must be complex. The quenching reaction of Xe with Kr_2F^* has a rate constant of 15×10^{-11} cm^3 s^{-1} and gives XeF(B, C); the mechanism also probably is displacement, but the KrXeF* molecules are formed above the dissociation limit for XeF(B, C)+Kr [11].

4. Conclusions

The two-photon, photo-assisted reaction between Kr and F_2 is a convenient way to prepare KrF(B, C) molecules for subsequent kinetic studies. A 7 ns pulse from a dye laser was used for this work, but the method is applicable for shorter times, if an appropriate laser is available. The coupled radiative lifetime for KrF(B, C) at 300 K is $\approx$25 ns, corresponding to an equilibrium constant of 4–5. The two-body quenching rate constants for a 300 K Boltzmann distribution of KrF(B, C) molecules are $(2.8\pm0.2)\times10^{-10}$, $(1.0\pm0.2)\times10^{-10}$, and $(4.0\pm0.1)\times10^{-12}$ cm^3 s^{-1} for F_2, Xe, and Kr, respectively. The three-body quenching rate by Kr was assigned as $(6.0\pm2.0)\times10^{-31}$ cm^6 s^{-1}. The Xe quenching reaction proceeds by a displacement reaction giving XeF(B, C). A comprehensive study of the two- and three-body quenching constants of KrF(B, C) and Kr_2F^* by molecules that are compatible with 1 Torr of F_2 is in progress. The general trend is for the quenching constants of KrF(B, C) to be larger than for Kr_2F^*, just as found [20] for XeCl(B, C) and Xe_2Cl^*. Such data should be useful for characterizing the quenching mechanisms of electronically excited states with a high degree of ionic character, as well aiding in the continued refinement of models for the KrF laser [21].

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